Networking Module 1 and 2

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1. **Consider a case, a folder has multiple files and how would copy it to destination machine path (Try using SCP, cp options in Linux)**
2. **Host a FTP and SFTP server and try PUT and GET operations.**
3. **Explore with Wireshark/TCP-dump/cisco packet tracer tools and learn about packets filters.**
4. **Understand linux utility commands like - Ping, arp (Understand each params from ifconfig output)**
5. **Understand what happens when duplicate IPs configured in a network.**
6. **Understand how to access remote system using (VNC viewer, Anydesk, teamviewer and remote desktop connections)**
7. **How to check your default gateway is reachable or not and understand about default gateway.**
8. **Check iwconfig/ifconfig to understand in detail about network interfaces (check about interface speed, MTU and other parameters)**
9. **Log in to your home router's web interface (usually at 192.168.1.1 or 192.168.0. l) and check the connected devices list.**
10. **Explain how a DHCP server assigns IP addresses to devices in your network.**
11. **Using a terminal, connect to a remote machine via SSH and telnet.**

Solutions:

1.

Copying a text file from Windows to Ubuntu using SCP

```
C:\Users\vimal>scp -r D:\EmbedUR\sampledata.txt vimal@172.20.10.2:/home/vimal/Desktop
vimal@172.20.10.2's password:
sampledata.txt                                                    100%   61    59.6KB/s   00:00
```

```
vimal@vimal-VirtualBox:~/Desktop$ ls
backup_manager.sh  file_analyzer.sh       listoffiles.txt  sample_text.txt
destination        file_analyzer.sh.save  output.txt       sample.txt
errors.log         input.txt              pract            source
extract_params.sh  input.txt.save         sampledata.txt   testdir
vimal@vimal-VirtualBox:~/Desktop$ cat sampledata.txt
Hi This Is Vimal, File transferred from Windows to VirtualBoxvimal@vimal-VirtualBox:~/D
esktop$
```

2.

```
C:\Users\vimal>ftp 172.20.10.2
Connected to 172.20.10.2.
220 (vsFTPd 3.0.5)
200 Always in UTF8 mode.
User (172.20.10.2:(none)): vimal
331 Please specify the password.
Password:

230 Login successful.
ftp> put D:\EmbedUR\ftp_sending.txt
200 PORT command successful. Consider using PASV.
150 Ok to send data.
226 Transfer complete.
ftp: 31 bytes sent in 0.00Seconds 31.00Kbytes/sec.
```

```
ftp> get /home/vimal/Desktop/ftptowindows.txt
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for /home/vimal/Desktop/ftptowindows.txt (39 bytes).
226 Transfer complete.
ftp: 39 bytes received in 0.00Seconds 39000.00Kbytes/sec.
```

```
sftp> put samplefile.txt
Uploading samplefile.txt to /home/vimal/samplefile.txt
samplefile.txt                              100%    0     0.0KB/s   00:00
sftp> get newfile.txt
Fetching /home/vimal/newfile.txt to newfile.txt
newfile.txt                                 100%  146    35.6KB/s   00:00
sftp> exit
```

3.

Wireshark is a tool for packet capturing and packet filtering
Wireshark has two types of filters.

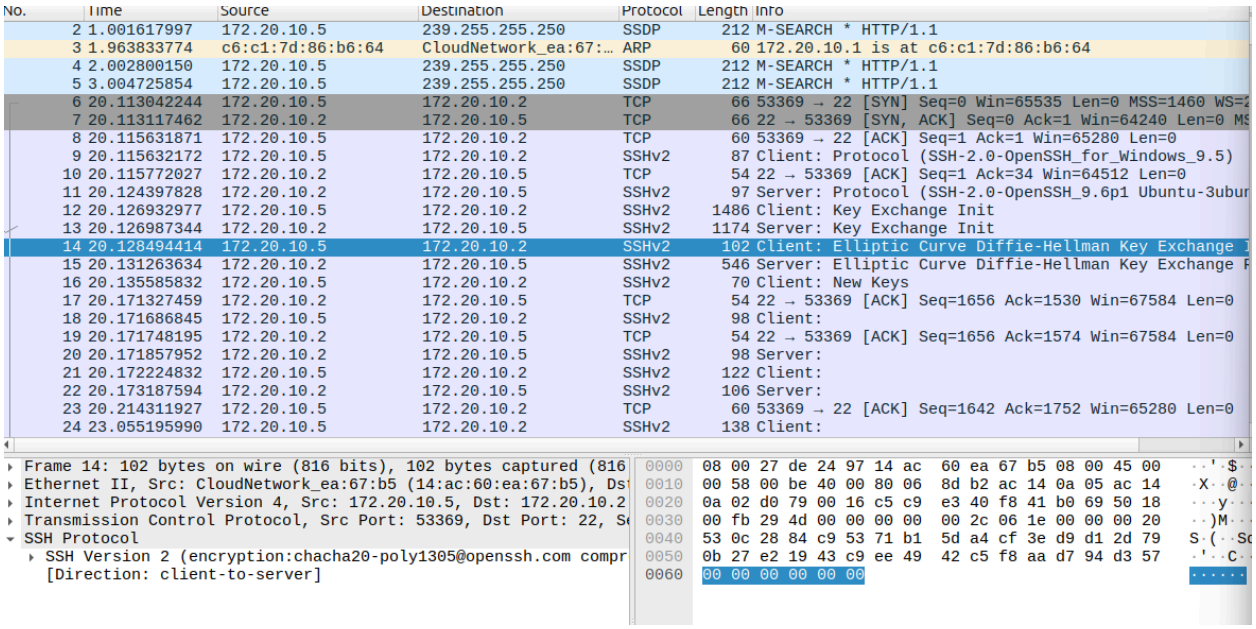
Capture filters are applied before capturing — only matching packets are recorded, which saves memory.
Display filters are applied after capturing — all packets are recorded but only matching ones are shown on screen.

While using FTP, it seems to show the password as plain text in one of the packet

Wireshark · Packet 20 · enp0s3

```
▸ Frame 20: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on int
▸ Ethernet II, Src: CloudNetwork_ea:67:b5 (14:ac:60:ea:67:b5), Dst: PCSSyste
▸ Internet Protocol Version 4, Src: 172.20.10.5, Dst: 172.20.10.2
▸ Transmission Control Protocol, Src Port: 53706, Dst Port: 21, Seq: 27, Ack
▾ File Transfer Protocol (FTP)
  ▾ PASS vimal\r\n
      Request command: PASS
      Request arg: vimal
  [Current working directory: ]
```

While using SFTP there isn't any packet exposing the username and password, everything is encrypted



No.	Time	Source	Destination	Protocol	Length	Info
2	1.001617997	172.20.10.5	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
3	1.963833774	c6:c1:7d:86:b6:64	CloudNetwork_ea:67:…	ARP	60	172.20.10.1 is at c6:c1:7d:86:b6:64
4	2.002800150	172.20.10.5	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
5	3.004725854	172.20.10.5	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
6	20.113042244	172.20.10.5	172.20.10.2	TCP	66	53369 → 22 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2
7	20.113117462	172.20.10.2	172.20.10.5	TCP	66	22 → 53369 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MS
8	20.115631871	172.20.10.5	172.20.10.2	TCP	60	53369 → 22 [ACK] Seq=1 Ack=1 Win=65280 Len=0
9	20.115632172	172.20.10.5	172.20.10.2	SSHv2	87	Client: Protocol (SSH-2.0-OpenSSH_for_Windows_9.5)
10	20.115772027	172.20.10.2	172.20.10.5	TCP	54	22 → 53369 [ACK] Seq=1 Ack=34 Win=64512 Len=0
11	20.124397828	172.20.10.2	172.20.10.5	SSHv2	97	Server: Protocol (SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubur
12	20.126932977	172.20.10.5	172.20.10.2	SSHv2	1486	Client: Key Exchange Init
13	20.126987344	172.20.10.2	172.20.10.5	SSHv2	1174	Server: Key Exchange Init
14	20.128494414	172.20.10.5	172.20.10.2	SSHv2	102	Client: Elliptic Curve Diffie-Hellman Key Exchange I
15	20.131263634	172.20.10.2	172.20.10.5	SSHv2	546	Server: Elliptic Curve Diffie-Hellman Key Exchange R
16	20.135585832	172.20.10.5	172.20.10.2	SSHv2	70	Client: New Keys
17	20.171327459	172.20.10.2	172.20.10.5	TCP	54	22 → 53369 [ACK] Seq=1656 Ack=1530 Win=67584 Len=0
18	20.171686845	172.20.10.5	172.20.10.2	SSHv2	98	Client:
19	20.171748195	172.20.10.2	172.20.10.5	TCP	54	22 → 53369 [ACK] Seq=1656 Ack=1574 Win=67584 Len=0
20	20.171857952	172.20.10.2	172.20.10.5	SSHv2	98	Server:
21	20.172224832	172.20.10.5	172.20.10.2	SSHv2	122	Client:
22	20.173187594	172.20.10.2	172.20.10.5	SSHv2	106	Server:
23	20.214311927	172.20.10.5	172.20.10.2	TCP	60	53369 → 22 [ACK] Seq=1642 Ack=1752 Win=65280 Len=0
24	23.055195990	172.20.10.5	172.20.10.2	SSHv2	138	Client:

```
▸ Frame 14: 102 bytes on wire (816 bits), 102 bytes captured (816
▸ Ethernet II, Src: CloudNetwork_ea:67:b5 (14:ac:60:ea:67:b5), Dst
▸ Internet Protocol Version 4, Src: 172.20.10.5, Dst: 172.20.10.2
▸ Transmission Control Protocol, Src Port: 53369, Dst Port: 22, Se
▾ SSH Protocol
  ▸ SSH Version 2 (encryption:chacha20-poly1305@openssh.com compr
    [Direction: client-to-server]
```

```
0000  08 00 27 de 24 97 14 ac  60 ea 67 b5 08 00 45 00   ··'·$··· ··g···E·
0010  00 58 00 be 40 00 80 06  8d b2 ac 14 0a 05 ac 14   ·X··@··· ····· ··
0020  0a 02 d0 79 00 16 c5 c9  e3 40 f8 41 b0 69 50 18   ···y···· ·@·A·iP·
0030  00 fb 29 4d 00 00 00 00  00 2c 06 1e 00 00 00 20   ··)M···· ·,····· 
0040  53 0c 28 84 c9 53 71 b1  5d a4 cf 3e d9 d1 2d 79   S·(··Sq· ]··>··-y
0050  0b 27 e2 19 43 c9 ee 49  42 c5 f8 aa d7 94 d3 57   ·'··C··I B·····W
0060  00 00 00 00 00 00                                  ····· ··
```

4.

PING & ARP

```
vimal@vimal-VirtualBox:~/Desktop$ arp
Address                  HWtype  HWaddress           Flags Mask            Iface
172.20.10.5              ether   14:ac:60:ea:67:b5   C                     enp0s3
_gateway                 ether   c6:c1:7d:86:b6:64   C                     enp0s3
vimal@vimal-VirtualBox:~/Desktop$ ping -c 3 www.google.com
PING www.google.com (2404:6800:4007:817::2004) 56 data bytes
64 bytes from maa05s16-in-x04.1e100.net (2404:6800:4007:817::2004): icmp_seq=1 ttl=116
time=31.9 ms
64 bytes from maa05s16-in-x04.1e100.net (2404:6800:4007:817::2004): icmp_seq=2 ttl=116
time=67.1 ms
64 bytes from maa05s16-in-x04.1e100.net (2404:6800:4007:817::2004): icmp_seq=3 ttl=116
time=41.9 ms

--- www.google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2190ms
rtt min/avg/max/mdev = 31.939/46.997/67.121/14.802 ms
```

<u>ARP</u>

Address — IP address of a device the machine has communicated with recently. Here, 172.20.10.5 is my host Windows and _gateway is the hotspot/router.

HWtype — Hardware type of the network interface. ether means both devices are using Ethernet (or virtual Ethernet in VirtualBox).

HWaddress — The MAC address (physical hardware ID) mapped to that IP. Here, 14:ac:60:ea:67:b5 belongs to the windows and c6:c1:7d:86:b6:64 belongs to gateway.

Flags — Status of the ARP entry. C means Complete — the mapping was successfully learned and is currently valid.

Iface — Which network interface learned this mapping. Here, both entries were learned through enp0s3.

PING

PING www.google.com (2404:6800:...) — The domain name that was pinged and the IP it resolved to. The resolution returned a Google IPv6 address because the hotspot supports IPv6.

56 data bytes — The size of each ping packet's payload. The default value of 56 bytes was used since no custom size was specified.

64 bytes from maa05s16... — Total reply size and the hostname of the server that responded. maa refers to the Chennai airport code, confirming that Google responded from their Chennai data center

icmp_seq — Sequence number of each packet, used to detect missing or out-of-order packets. Sequences 1, 2, and 3 all returned, confirming no packets were dropped.

ttl=116 — Time To Live, which decrements by 1 at each router hop. A starting value of 128 minus the received value of 116 means the packet traversed 12 routers to reach Google.

time — Round trip time for each individual packet. The three packets took 31.9ms, 67.1ms, and 41.9ms respectively, showing slight variation typical of a mobile hotspot connection.

0% packet loss — Percentage of packets sent that received no reply. A value of 0% confirms all 3 packets returned successfully, indicating a healthy connection.

rtt min/avg/max — The fastest, average, and slowest response times recorded across all packets. Values of min=31.9ms, avg=47ms, max=67.1ms are within normal range for a mobile hotspot.

mdev=14.802ms — Mean deviation, representing how much latency varies between packets. A value of 14.8ms indicates mild inconsistency, which is expected on a mobile hotspot. A stable wired connection would typically show under 1ms.

5.

A duplicate IP occurs when two devices in the same Layer 2 broadcast domain are configured with the same IPv4 address.

Device A → 10.0.0.5

Device B → 10.0.0.5

This is an invalid configuration and causes ARP instability.

Since ARP resolves IP addresses to MAC addresses, when Device C sends an ARP request for 10.0.0.5, both Device A and B respond with their respective MAC addresses.

As a result, traffic gets delivered to either device randomly, leading to intermittent connectivity, flapping communication, and dropped sessions.

The ARP table continuously flips the MAC address mapped to the same IP, making communication unreliable.

6.

1. VNC — Virtual Network Computing

VNC uses the RFB (Remote Framebuffer Protocol) where the server captures screen pixels and sends framebuffer updates to the client, while the client sends back keyboard and mouse inputs. It does not handle NAT traversal automatically, so either port forwarding or SSH tunneling is required for internet access. Default port is TCP 5900. Popular implementations include RealVNC and TightVNC.

2. RDP — Remote Desktop Protocol

Microsoft's native remote desktop protocol built into Windows. Instead of transmitting raw pixels like VNC, RDP sends drawing commands making it more bandwidth efficient. It also supports device redirection for printers, drives, and clipboard. Default port is TCP 3389.

3. TeamViewer

TeamViewer routes connections through cloud relay servers, so no port forwarding is needed. Both machines connect outbound to TeamViewer's servers, exchange IDs, and an encrypted tunnel is established either peer-to-peer or via relay. Primary port is TCP 5938, with fallback to 443 or 80.

4. AnyDesk

Similar architecture to TeamViewer with automatic NAT traversal and relay servers. Uses a proprietary DeskRT codec for efficient screen compression, resulting in low-latency sessions even on slower connections. Default outbound ports are TCP 80, 443, and 6568.

7.

Finding the default gateway and using ping to check reachable or not

```
vimal@vimal-VirtualBox:~/Desktop$ ip route | grep default
default via 172.20.10.1 dev enp0s3 proto dhcp src 172.20.10.2 metric 100
vimal@vimal-VirtualBox:~/Desktop$ ping -c 3 172.20.10.1
PING 172.20.10.1 (172.20.10.1) 56(84) bytes of data.
64 bytes from 172.20.10.1: icmp_seq=1 ttl=64 time=4.11 ms
64 bytes from 172.20.10.1: icmp_seq=2 ttl=64 time=4.95 ms
64 bytes from 172.20.10.1: icmp_seq=3 ttl=64 time=4.09 ms

--- 172.20.10.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2015ms
rtt min/avg/max/mdev = 4.093/4.383/4.947/0.398 ms
```

A default gateway is the router IP address used to forward traffic to other networks.
It is used when the destination IP is outside the local subnet, and acts as the exit point from your local network to external networks.

Example:

IP Address: 172.20.10.2 — Ubuntu VM's IP

Subnet: 172.20.10.0/28 — local network range

Broadcast: 172.20.10.15 — last address in your subnet

Default Gateway: 172.20.10.1 — router's IP *(first usable address in the subnet)*

This means any traffic not within 172.20.10.0/28 is sent to 172.20.10.1 to be routed out to the internet.

8.

```
vimal@vimal-VirtualBox:~/Desktop$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.20.10.2  netmask 255.255.255.240  broadcast 172.20.10.15
        inet6 2401:4900:6332:5662:5f93:859:4a2a:a7fe  prefixlen 64  scopeid 0x0<
global>
        inet6 fe80::a00:27ff:fef5:6817  prefixlen 64  scopeid 0x20<link>
        inet6 2401:4900:6332:5662:a00:27ff:fef5:6817  prefixlen 64  scopeid 0x0<
global>
        ether 08:00:27:f5:68:17  txqueuelen 1000  (Ethernet)
        RX packets 189  bytes 33322 (33.3 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 254  bytes 34243 (34.2 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 85  bytes 9343 (9.3 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 85  bytes 9343 (9.3 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

1. MTU (Maximum Transmission Unit)

The largest size of a packet that can be sent over this interface in one go

enp0s3 → MTU is 1500 (standard Ethernet size).

If a packet is larger than this, it is fragmented.

lo → MTU is 65536. Since it is internal to the CPU/RAM, it can handle very large packets without fragmentation.

2. Type (Ethernet / Loopback)

enp0s3 : Ethernet (physical or virtual network card).

lo : Loopback (internal system communication interface).

3. Flags

UP: The interface is software-activated.

RUNNING: The interface is operational (connected/active).

BROADCAST: Can send messages to all devices in the local network.

MULTICAST: Can send data to a selected group of devices simultaneously.

LOOPBACK (lo only): Internal interface used by the system to communicate with itself.

4. TX/RX Statistics (Traffic Analysis)

RX (Receive): Incoming traffic.

Packets: Total number of received packets.

Bytes: Total amount of data received.

Errors: Transmission errors (should ideally be 0).

Dropped: Packets discarded (usually due to buffer overflow).

Overruns: CPU could not process packets fast enough.

Frame: Frame alignment errors.

TX (Transmit): Outgoing traffic.

Packets: Total number of transmitted packets.

Bytes: Total amount of data sent.

Errors: Transmission errors.

Dropped: Packets not sent successfully.

Overruns: Transmission buffer overflow.

Carrier: Carrier signal problems.

Collisions: Packet collisions on the network (common in half-duplex networks).

5. Scope (IPv6 Addresses)

Scopeid 0x20 <link>:

Link-local IPv6 address. Valid only within the local network segment. Cannot access the internet.

Scopeid 0x0 <global>:

Global IPv6 address. Can be used for communication outside the local network (internet-reachable in theory).

9.

I am using a mobile hotspot for the internet. Since a mobile hotspot is being used instead of a home router, there is no web interface like 192.168.1.1 to log into. The connected devices list can be viewed directly on the phone under Settings - Hotspot - Connected Devices.

But command like nmap can be used to discover devices on a network in linux.

```
vimal@vimal-VirtualBox:~/Desktop$ sudo nmap -sn 172.20.10.0/28
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-03 20:23 IST
Nmap scan report for _gateway (172.20.10.1)
Host is up (0.0053s latency).
MAC Address: C6:C1:7D:86:B6:64 (Unknown)
Nmap scan report for 172.20.10.5
Host is up (0.00031s latency).
MAC Address: 14:AC:60:EA:67:B5 (Cloud Network Technology Singapore PTE.)
Nmap scan report for vimal-VirtualBox (172.20.10.2)
Host is up.
Nmap done: 16 IP addresses (3 hosts up) scanned in 1.73 seconds
```

10.

DHCP — Dynamic Host Configuration Protocol

Every device in a network requires an IP address, subnet mask, default gateway, and DNS server to communicate. Manually configuring these for every device is not scalable, which is where DHCP comes in. DHCP automates IP configuration using a client-server model and operates at the application layer over UDP, using port 67 on the server and port 68 on the client.

DORA Process

Step 1 — DHCP Discover

When a device boots without an IP address, it broadcasts a discovery packet across the network to locate a DHCP server.

Source IP : 0.0.0.0

Destination IP : 255.255.255.255

UDP 68 - 67

Step 2 : DHCP Offer

The DHCP server receives the request and checks its available IP pool, the client's MAC address, and the lease database. It then responds with an offer containing an IP

address, subnet mask, default gateway, DNS server, and lease time. At this stage the IP is not permanently assigned yet.

Step 3 : DHCP Request

The client may receive offers from multiple DHCP servers. It selects one and broadcasts an acceptance message. This broadcast also informs the other DHCP servers to withdraw their offers.

Step 4 : DHCP Acknowledge

The selected server confirms the IP assignment and starts the lease timer, completing the process.

Lease Mechanism

IP assignments are not permanent. When the lease reaches 50% of its duration, the client sends a renewal request directly to the server. If the server responds with an acknowledgment, the lease is extended. If no response is received, the client eventually releases the IP and restarts the DORA process from the beginning.

```
vimal@vimal-VirtualBox:~/Desktop$ ps aux | grep -E 'NetworkManager|systemd-networkd|dhclient'
root         851  0.0  0.2 336320 19960 ?        Ssl  15:05   0:00 /usr/sbin/NetworkManager --no-daemon
vimal       4843  0.0  0.0   9296  2408 pts/0    S+   15:46   0:00 grep --color=auto -E NetworkManager|systemd-networ
kd|dhclient
vimal@vimal-VirtualBox:~/Desktop$ nmcli device show enp0s3
GENERAL.DEVICE:                         enp0s3
GENERAL.TYPE:                           ethernet
GENERAL.HWADDR:                         08:00:27:F5:68:17
GENERAL.MTU:                            1500
GENERAL.STATE:                          100 (connected)
GENERAL.CONNECTION:                     netplan-enp0s3
GENERAL.CON-PATH:                       /org/freedesktop/NetworkManager/ActiveConnecti>
WIRED-PROPERTIES.CARRIER:               on
IP4.ADDRESS[1]:                         172.20.10.2/28
IP4.GATEWAY:                            172.20.10.1
IP4.ROUTE[1]:                           dst = 172.20.10.0/28, nh = 0.0.0.0, mt = 100
IP4.ROUTE[2]:                           dst = 0.0.0.0/0, nh = 172.20.10.1, mt = 100
IP4.DNS[1]:                             172.20.10.1
IP6.ADDRESS[1]:                         2401:4900:6332:5662:a00:27ff:fef5:6817/64
IP6.ADDRESS[2]:                         2401:4900:6332:5662:5f93:859:4a2a:a7fe/64
IP6.ADDRESS[3]:                         fe80::a00:27ff:fef5:6817/64
IP6.GATEWAY:                            fe80::c4c1:7dff:fe86:b664
IP6.ROUTE[1]:                           dst = fe80::/64, nh = ::, mt = 256
IP6.ROUTE[2]:                           dst = ::/0, nh = fe80::c4c1:7dff:fe86:b664, mt>
IP6.ROUTE[3]:                           dst = 2401:4900:6332:5662::/64, nh = ::, mt = >
vimal@vimal-VirtualBox:~/Desktop$ sudo nmcli device disconnect enp0s3
[sudo] password for vimal:
Device 'enp0s3' successfully disconnected.
vimal@vimal-VirtualBox:~/Desktop$ sudo nmcli device connect enp0s3
Device 'enp0s3' successfully activated with '1eef7e45-3b9d-3043-bee3-fc5925c90273'.
vimal@vimal-VirtualBox:~/Desktop$ sudo tcpdump -i enp0s3 port 67 or port 68 -n
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
15:50:33.673118 IP 0.0.0.0.68 > 255.255.255.255.67: BOOTP/DHCP, Request from 08:00:27:f5:68:17, length 300
15:50:33.685817 IP 172.20.10.1.67 > 255.255.255.255.68: BOOTP/DHCP, Reply, length 300
```

11.

Starting the server and verifying

```
vimal@vimal-VirtualBox:~/Desktop$ sudo systemctl start ssh
[sudo] password for vimal:
vimal@vimal-VirtualBox:~/Desktop$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enab>
     Active: active (running) since Tue 2026-03-03 19:36:21 IST; 15min ago
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 1218 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1233 (sshd)
      Tasks: 1 (limit: 9434)
     Memory: 2.1M (peak: 2.7M)
        CPU: 37ms
     CGroup: /system.slice/ssh.service
             └─1233 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 03 19:36:21 vimal-VirtualBox systemd[1]: Starting ssh.service - OpenBSD Sec>
Mar 03 19:36:21 vimal-VirtualBox sshd[1233]: Server listening on 0.0.0.0 port 2>
Mar 03 19:36:21 vimal-VirtualBox sshd[1233]: Server listening on :: port 22.
Mar 03 19:36:21 vimal-VirtualBox systemd[1]: Started ssh.service - OpenBSD Secu>
```

Connecting to remote host

```
C:\Users\vimal>ssh vimal@172.20.10.2
vimal@172.20.10.2's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.17.0-14-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

273 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

8 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Mon Mar  2 23:06:30 2026 from 172.20.10.5
```

Starting the telnet server

```
vimal@vimal-VirtualBox:~/Desktop$ sudo systemctl start inetutils-inetd
vimal@vimal-VirtualBox:~/Desktop$ sudo systemctl status inetutils-inetd
● inetutils-inetd.service - GNU Network Utilities internet superserver
     Loaded: loaded (/usr/lib/systemd/system/inetutils-inetd.service; enabled; preset:>
     Active: active (running) since Tue 2026-03-03 20:18:55 IST; 5s ago
       Docs: man:inetutils-inetd(8)
             https://www.gnu.org/software/inetutils/manual/
    Process: 5717 ExecCondition=grep -qr ^[0-9A-Za-z/] /etc/inetd.conf /etc/inetd.d/ (>
   Main PID: 5719 (inetutils-inetd)
      Tasks: 1 (limit: 9434)
     Memory: 212.0K (peak: 1.6M)
        CPU: 15ms
     CGroup: /system.slice/inetutils-inetd.service
             └─5719 /usr/sbin/inetutils-inetd --foreground

Mar 03 20:18:55 vimal-VirtualBox systemd[1]: Starting inetutils-inetd.service - GNU Ne>
Mar 03 20:18:55 vimal-VirtualBox systemd[1]: Started inetutils-inetd.service - GNU Net>
lines 1-15/15 (END)
```

Accessing the host

```
C:\Users\vimal>telnet 172.20.10.2
```

```
Linux 6.17.0-14-generic (vimal-VirtualBox) (pts/2)

vimal-VirtualBox login: vimal

Password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.17.0-14-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

273 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

8 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm
```